MILITARY SPECIFICATION

CALCULATOR, NUCLEAR YIELD, M4A1

1. SCOPE

1.1 This specification covers one type of nuclear yield calculator. The calculator is a circular type slide rule consisting of three concentric disks containing the applicable scales.

2. APPLICABLE DOCUMENTS

2.1 Government documents. The following documents of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 - Marking for Shipment and Storage.

DRAWINGS

US ARMY MUNITIONS COMMAND

EDGEWOOD ARSENAL

DL 5-86-36 - Calculator, Nuclear Yield, M4A1.

(Copies of specifications, standards, drawings and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

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UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification

(Application for copies of these ratings, rules and regulations should be addressed to Uniform Classification Committee, 202 Union Station, 516 West Jackson Boulevard, Chicago, Illinois 60606.)

3. REQUIREMENTS

3.1 Materials and components.

3.1.1 Materials. All materials cited on Drawing DL5-86-36 or on the subsidiary drawings shall conform to the specifications listed thereon, or to the specific characteristics set forth on the drawings. The ink used for marking and lettering shall be compatible with the vinyl chloride material. The printing shall be legible, nonfading, and durable.

3.1.2 Components. All components of the calculator shall conform to the specifications and drawings listed on Drawing DL 5-86-36 and subsidiary drawings.

3.2 Manufacture and assembly. The calculator shall be manufactured and assembled in accordance with Drawing C5-86-36.

3.3 Functioning. The calculator, when tested in accordance with 4.4.4, shall be capable of solving the problems for yield using the data listed in table I.

Table I. Problem data.

Scale	Data		
	Problem 1	Problem 2	Problem 3
Distance to ground zero	---	---	100 kilometers
Flash to bang time	65 secs	20 secs	---
Elevation angle	800 mils	750 mils	---
Cloud width	---	---	85 mils
Yield - Cloud top	1500 ± 200 kt	---	---
Yield - Cloud bottom	---	10 ± 1.0 kt	---
Yield	---	---	100 ± 2 kt

3.4 Preproduction. Prior to the start of regular production, a preproduction sample of M4A1 calculators shall be produced in accordance with this specification for examination and test (see 4.3).

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3.5 Workmanship. The calculator shall be free from imperfections (pits, bumps, cracks, flash, abrasion, blisters, holes, or porous areas) and contamination (dirt, grease, oil or foreign matter). All surfaces shall be smooth and laminations shall be uniform without unsealed areas.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection.

4.1.1 Supplier's responsibility. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to specified requirements.

4.1.2 Objective evidence. The supplier shall provide objective evidence acceptable to the contracting officer that the requirements of 3.1 and section 5 for which specific inspection has not been provided in this specification have been satisfied.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

(a) Preproduction inspection (see 4.3).
(b) Quality conformance inspection (see 4.4).

4.3 Preproduction inspection.

4.3.1 Sample. A preproduction sample of five completely assembled M4A1 calculators and five unassembled M4A1 calculators shall be manufactured using the same methods, materials, equipment and processes as will be used during regular production.

4.3.2 Examination and nondestructive test. The components of the five unassembled calculators shall be examined for all the requirements of the drawings. The five assembled calculators shall be examined in accordance with the classification of defects (4.4.3.2).

4.3.3 Acceptance/rejection criteria. The preproduction sample shall comply with the examinations and tests specified in 4.3.2 to be acceptable. The supplier shall obtain written approval from the contracting officer prior to proceeding with regular production.

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4.4 Quality conformance inspection.

4.4.1 Lotting. A lot shall consist of the M4A1 calculators produced by one manufacturer, at one plant, from the same materials, under essentially the same manufacturing conditions.

4.4.2 Sampling for examination and nondestructive tests. Sampling shall be conducted in accordance with MIL-STD-105. For major defect 101, level S-2 shall be used.

4.4.3 Inspection procedures.

4.4.3.1 Examination and nondestructive tests. Sample calculators shall be examined and tested in accordance with the classification of defects and MIL-STD-105.

4.4.3.2 Classification of defects.

(a) Calculator, nuclear yield, M4A1 (Dwg C5-86-36).

Categories		Defects	Acceptance standards
Critical:		None defined	
Major:	AQL	1.0 percent defective	
101		Functioning	4.4.4
	AQL	2.5 percent defective	
102		Component missing, incorrect or incorrectly assembled	
103		Disks not free to turn easily	
104		Marking or lettering missing, incorrect, or illegible, or color incorrect	
105		Rivet holes in disks incorrectly located	
106		Workmanship (see 3.5)	

4.4.4 Functioning test. Test shall be conducted as follows:

(a) Problem 1. Align 65 seconds on the "Flash-to-Bang Time" scale on the front inner disk with 800 mils on the "Stabilized Cloud Bottom or Top Elevation Angle" scale on the outer disk. Read answer on the "Yield Stabilized Cloud Top" scale of the outer disk under the pointer of the inner disk.

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(b) Problem 2. Align 20 seconds on the "Flash-to-Bang-Time" scale on the front inner disk with 750 mils on the "Stabilized Cloud Bottom or Top Elevation Angle" scale on the outer disk. Read answer on the "Yield Stabilized Cloud Bottom" scale of the outer disk under the pointer of the inner disk.

(c) Problem 3. Align 100 kilometers in the "Distance to Ground Zero" scale on the back inner disk with 85 mils on the "Observed Cloud Width at Bang Time" scale on the outer disk. Read answer on the "Yield" scale of the outer disk under the pointer of the inner disk.

5. PREPARATION FOR DELIVERY

5.1 Packaging, level C. The calculators shall be packaged to provide adequate protection against deterioration and damage from the supply source to the first receiving activity for immediate use.

5.2 Packing, level C. Calculators, packaged as specified in 5.1, shall be packed to insure carrier acceptance and safe delivery to the first domestic destination. Containers shall be in compliance with Uniform Freight Classification rules or the regulations of other common carriers applicable to the mode of transportation.

5.3 Marking. In addition to any special marking required by the contract or order, unit packages and shipping containers shall be marked in accordance with MIL-STD-129.

6. NOTES

6.1 Intended use. This calculator is intended for use in calculating the nuclear yield from an atomic burst.

6.2 Ordering data. Procurement documents should specify the following:

(a) Title, number and date of this specification.
(b) Preproduction:

(1) Time allowed for supplier submission of samples for Government test and evaluation after award of contract.
(2) Name and address of test facility and shipping instructions when testing is performed by the Government.
(3) Time allowed for the Government to notify the supplier whether or not to proceed with production.

Custodians:

Army - MU

Preparing activity:

Army - MU(EA)

Project No. 6665-A236

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE | CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1 HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A GIVE PARAGRAPH NUMBER AND WORDING

B RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2 COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3 IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4 REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)* | DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64 WHICH MAY BE USED

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